

Candidate Recommendation System

Matching candidate CVs to job openings automatically, using TF-IDF vectorization and cosine similarity.

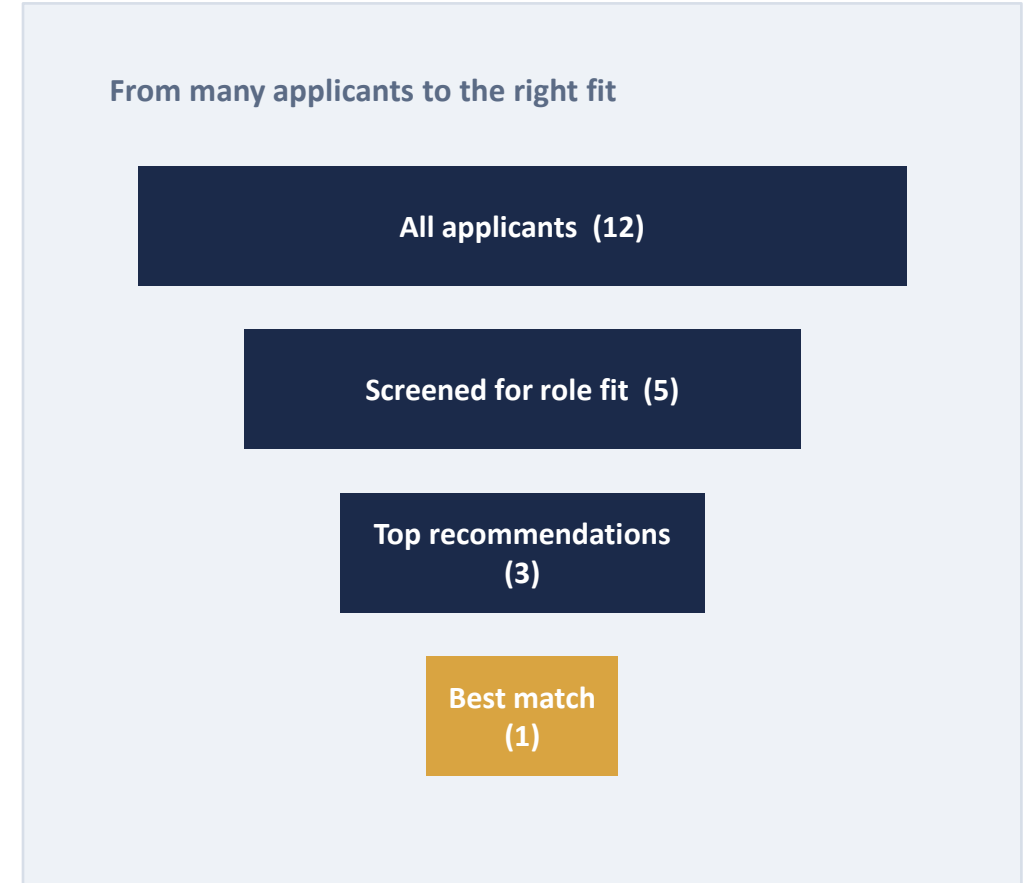
THE PROBLEM

Manual resume screening doesn't scale

Recruiters at a staffing firm receive far more applications than they can carefully read. Screening by hand is slow, inconsistent between reviewers, and easy to bias toward whichever resume happens to be read first.

Our objective

- Automatically compare candidate CVs against job descriptions
- Score how well each candidate fits a role, using their skills, experience and education
- Rank candidates so recruiters can focus on the strongest matches first



A sample of 12 candidates across 4 real job roles

We drew our candidates and job postings from a public resume-matching dataset (Kaggle), then cleaned and sampled it down to a manageable set for this project: 12 candidate CVs matched against 4 distinct job openings.

Senior Software Engineer

3 candidates

Technical support, deployment, mentorship

HR Officer

3 candidates

HR management, policy, employee relations

Civil Engineer

3 candidates

AutoCAD, design review, site supervision

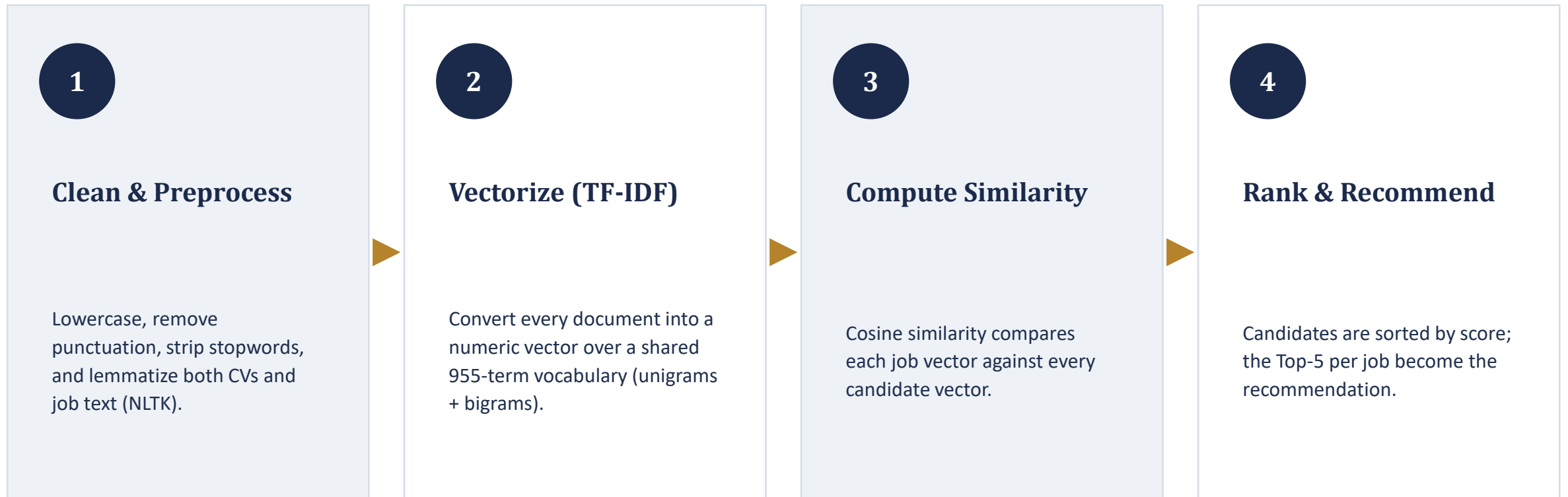
Business Development Executive

3 candidates

Sourcing, vendor & buyer relations

Each CV includes career objective, skills, education, and work history. Each job posting lists its own requirements, so the system has independent text on both sides to compare.

A four-stage pipeline, start to finish



We also tried Word2Vec as a comparison method, but with only 16 short documents there wasn't enough text to train reliable word embeddings — so TF-IDF is the method used for the final recommendations.

Demo — Business Development Executive

For each job posting, the system ranks all 12 candidates by cosine similarity and returns the Top-5. Here is the actual output for one of our four jobs.

Rank	Candidate	Similarity Score	
1	candidate_08	0.0658	TOP MATCH
2	candidate_11	0.0246	
3	candidate_10	0.0153	
4	candidate_01	0.0118	
5	candidate_04	0.0094	

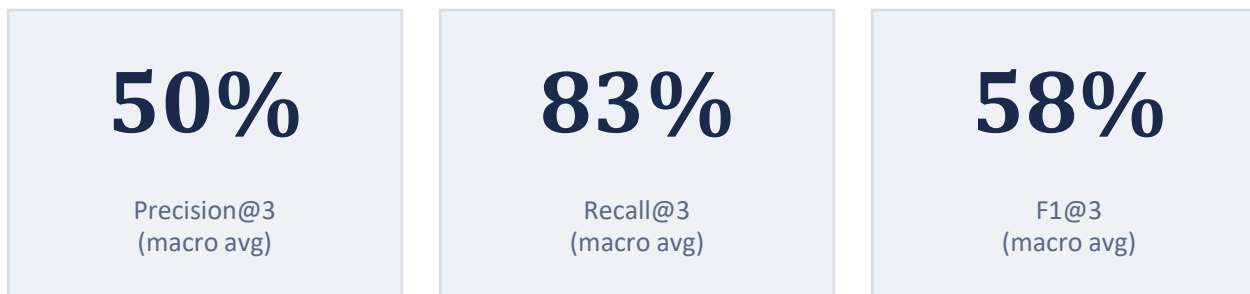
Why candidate_08 ranks first

candidate_08's CV literally contains “business development executive,” plus brand development, sales, account management and negotiation — direct word overlap with the job posting drives the highest cosine similarity of the group.

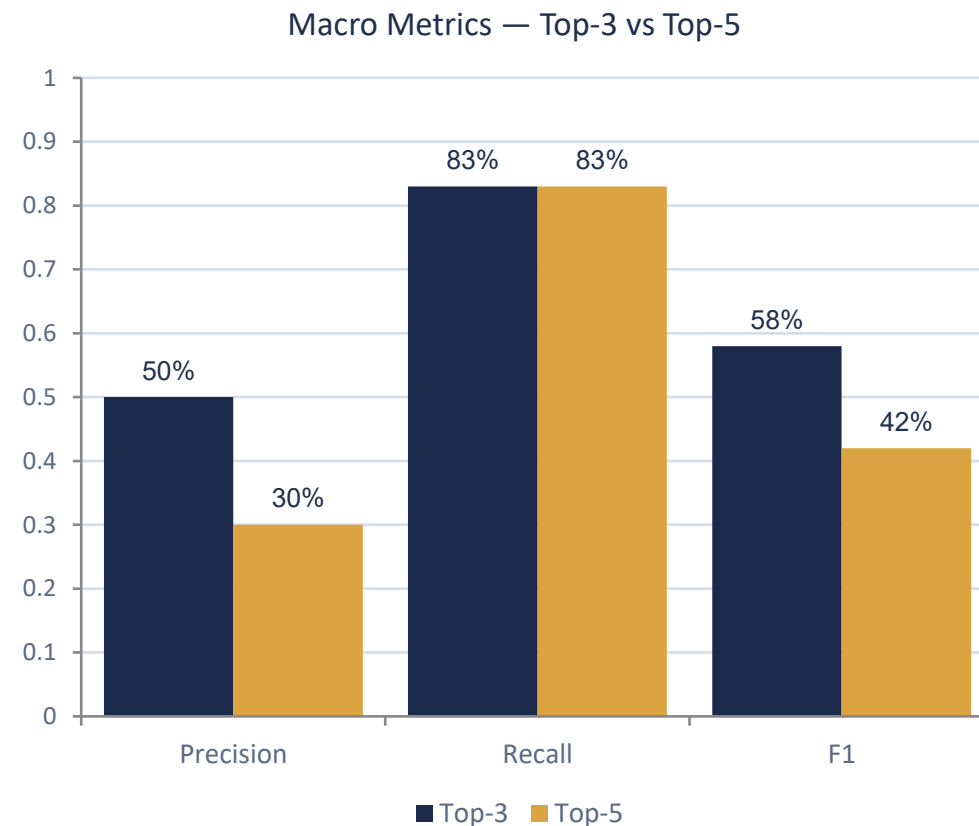
Similarity scores are small in absolute terms (TF-IDF vectors are sparse) — what matters is the ranking they produce, not the raw number.

How well does the system actually perform?

The dataset's own match scores didn't hold up under review — in several cases the highest-scored candidate for a job had no relevant skills at all. So as a team we manually judged all 48 candidate-job pairs ourselves and built our own answer key (8 genuinely relevant pairs) to evaluate against.



Every job's single best candidate was ranked in our system's Top-3 for that role.



Strong hits, and one honest miss

WORKED WELL

- candidate_06 (mechanical engineer, AutoCAD experience) ranked #1 for Civil Engineer — the only real cross-domain match in our data, and the system found it.
- candidate_08 ranked #1 for Business Development Executive by a wide margin — the clearest true match in the whole dataset.
- In every job, the strongest known match landed inside the Top-3 recommendations.

A REAL LIMITATION

candidate_02 had the most genuinely relevant software-development skills for Senior Software Engineer (APIs, JavaScript, REST) — but ranked only 6th.

TF-IDF only matches literal shared words. The job posting used generic terms like “technical support” and “monitoring”, which don't overlap with candidate_02's specific tool names — even though a human reviewer would clearly see the fit.

This is why we recommend semantic embeddings (e.g. Sentence-BERT) as the next step — see Conclusion.

CONCLUSION

A working recommendation engine — and a clear next step

What we delivered

- A full pipeline: cleaning → TF-IDF vectorization → cosine similarity → ranked Top-5 recommendations
- A team-verified 48-pair evaluation set, built after the original data proved unreliable
- Precision, Recall and F1 measured at Top-3 and Top-5 for every job

Where we'd take it next

- Sentence-BERT embeddings, to catch semantic matches TF-IDF misses
- A larger candidate and job pool for more reliable metrics
- Separate, weighted scoring for skills, experience and education instead of one blended score

Thank you.